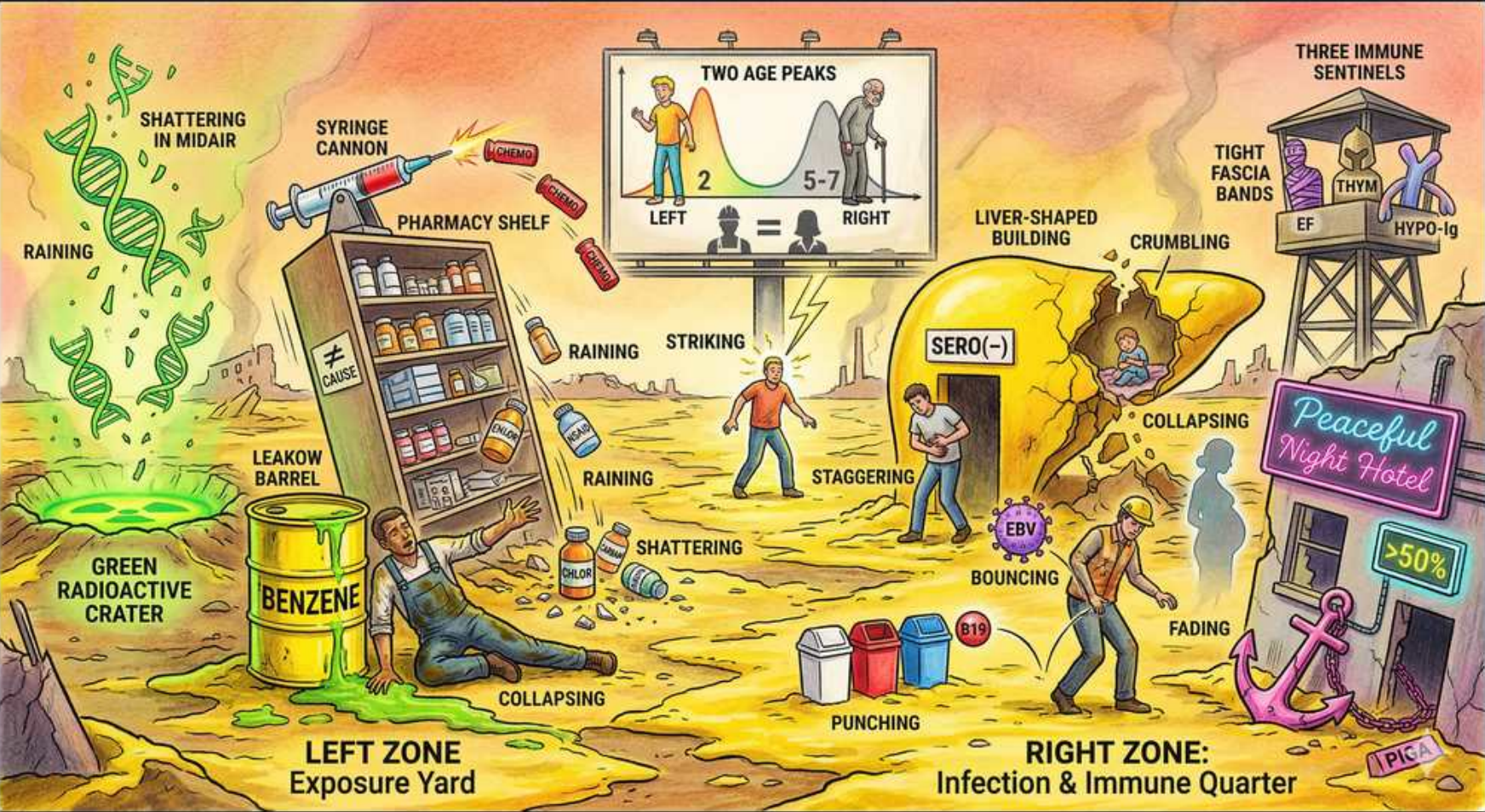


Bone Marrow Failure — The Aplastic Wasteland (AA Etiology)



1. Two age peaks

WHAT YOU SEE

Billboard graph with a young figure (left) and elderly figure (right), peaks 2/57+

WHAT IT ENCODES

Acquired aplastic anemia has a BIMODAL age distribution — a first peak in adolescents/young adults (~15–25 yr) and a second, larger peak in older adults (>60 yr). Overall it is a rare disorder (~2 per million/year in the West, several-fold higher in East Asia). It presents as pancytopenia from an empty (hypocellular) marrow.

BOARD TRAP

Don't picture a single middle-age peak — the classic teaching is two peaks (young AND old). Also distinguish ACQUIRED AA from INHERITED bone-marrow-failure syndromes (Fanconi anemia, dyskeratosis congenita, Shwachman-Diamond) which present in children and have congenital stigmata.

2. Idiopathic/autoimmune

WHAT YOU SEE

LEFT = RIGHT equals sign under the billboard

WHAT IT ENCODES

The MOST COMMON cause is IDIOPATHIC (no identifiable exposure) — the majority of cases. The underlying mechanism even in 'idiopathic' disease is immune: autoreactive cytotoxic T cells destroy hematopoietic stem cells, which is why immunosuppression works.

BOARD TRAP

Boards want 'idiopathic/immune-mediated' as the #1 cause. The named exposures (drugs, toxins, radiation, viruses) are the classic minority — high-yield to recognize but NOT the most frequent etiology.

3. Pharmacy shelf

WHAT YOU SEE

Tall PHARMACY shelf stocked with drug bottles

WHAT IT ENCODES

Drugs are a classic IDIOSYNCRATIC (non-dose-dependent) cause. CHLORAMPHENICOL is the board archetype. Others: sulfonamides, NSAIDs (phenylbutazone), antithyroid drugs (propylthiouracil, methimazole/carbimazole), antiepileptics (carbamazepine, phenytoin), gold, and ticlopidine.

BOARD TRAP

Separate IDIOSYNCRATIC drug aplasia (unpredictable, any dose — e.g., chloramphenicol) from DOSE-DEPENDENT marrow suppression of cytotoxic chemo (predictable, reversible). Chloramphenicol is famous for BOTH: a reversible dose-related suppression AND a rare fatal idiosyncratic aplasia.

4. Syringe cannon

WHAT YOU SEE

Syringe cannon labeled firing into the air

WHAT IT ENCODES

Cytotoxic CHEMOTHERAPY and other dose-dependent agents directly suppress marrow stem cells — this is expected, predictable, and usually reversible after the drug clears.

BOARD TRAP

This dose-dependent suppression is distinct from true idiosyncratic aplastic anemia; it is anticipated marrow toxicity, not an immune idiosyncratic reaction.

5. Benzene barrel

WHAT YOU SEE

Leaking BENZENE barrel beside a collapsed worker, LEAKY BARREL

WHAT IT ENCODES

BENZENE is the classic industrial/occupational toxin causing aplastic anemia (and later myelodysplasia/AML). Exposure is dose-related (paint, solvents, petrochemicals, glue, rubber, cigarette smoke).

BOARD TRAP

Benzene links AA to later clonal evolution — chronic exposure can cause aplastic anemia AND predisposes to MDS/acute myeloid leukemia. Recognize the occupational-history clue on the exam.

6. Green radioactive crater

WHAT YOU SEE

Green glowing radioactive crater at far left

WHAT IT ENCODES

IONIZING RADIATION damages hematopoietic stem cell DNA and causes dose-dependent marrow aplasia (whole-body irradiation, radiotherapy, nuclear accidents). Like benzene, it also carries later leukemia/MDS risk.

BOARD TRAP

Radiation marrow injury is dose-dependent and predictable — contrast with the idiosyncratic, dose-independent drug reactions on the pharmacy shelf.

7. Shattering DNA

WHAT YOU SEE

Green DNA helix shattering in midair, raining fragments

WHAT IT ENCODES

Common final pathway: these exposures (chemicals, radiation, some drugs) damage hematopoietic stem-cell DNA, depleting the stem-cell pool and producing a hypocellular 'wasteland' marrow.

BOARD TRAP

DNA-damage/short-telomere biology overlaps with INHERITED syndromes — short telomeres (dyskeratosis congenita) and Fanconi-pathway DNA-repair defects can masquerade as acquired AA, so chromosome-breakage and telomere testing are done in young patients.

8. Hepatitis liver

WHAT YOU SEE

Crumbling LIVER-SHAPED building labeled SERO(-)

WHAT IT ENCODES

POST-HEPATITIS aplastic anemia: severe AA can follow an episode of acute hepatitis, characteristically SERONEGATIVE (non-A, non-B, non-C) hepatitis, classically in a young male, typically 1–3 months after the hepatitis resolves.

BOARD TRAP

The hepatitis is usually NOT one of the identifiable viral serotypes (A/B/C) — it is seronegative. This is an immune-mediated phenomenon and responds to immunosuppression like other AA.

9. Parvovirus / EBV

WHAT YOU SEE

Bouncing ball labeled EBV near the infection quarter

WHAT IT ENCODES

Viral triggers of marrow failure: EBV, CMV, and HIV can cause aplasia. PARVOVIRUS B19 is the classic erythroid-specific suppressor causing PURE RED CELL APLASIA (transient aplastic crisis in chronic hemolysis; chronic PRCA in immunocompromised).

BOARD TRAP

Don't confuse the two: Parvovirus B19 → PURE RED CELL APLASIA (isolated anemia, giant pronormoblasts), NOT pancytopenic aplastic anemia. True pancytopenic AA points more to EBV/seronegative hepatitis/idiopathic immune disease.

10. Immune sentinels tower

WHAT YOU SEE

Watchtower with THREE IMMUNE SENTINELS and tight fascia bands

WHAT IT ENCODES

Core PATHOGENESIS: autoreactive cytotoxic CD8 T cells, driven by IFN- γ and TNF- α , attack and destroy hematopoietic stem/progenitor cells via Fas-mediated apoptosis. This immune mechanism is why ATG + cyclosporine (immunosuppression) restores hematopoiesis.

BOARD TRAP

AA is NOT primarily a stem-cell-intrinsic defect in most acquired cases — it is an IMMUNE attack on the marrow. That distinction underlies the entire treatment strategy.

11. Thymoma flag

WHAT YOU SEE

THYM flag with HYPO-Ig on the tower

WHAT IT ENCODES

Immune-dysregulation associations: THYMOMA (and hypogammaglobulinemia) is classically tied to PURE RED CELL APLASIA, reflecting T-cell-mediated marrow injury; the same autoimmune theme underlies aplastic anemia.

BOARD TRAP

Thymoma's strongest hematologic association is PRCA (look for thymoma + isolated anemia → image the mediastinum / consider thymectomy), not full pancytopenic AA — a common distractor.

12. Peaceful Night Hotel >50%

WHAT YOU SEE

Pink PEACEFUL NIGHT HOTEL sign reading >50%

WHAT IT ENCODES

PNH OVERLAP: a small paroxysmal nocturnal hemoglobinuria clone is detectable by flow cytometry in well over half of acquired AA patients at diagnosis. Screen every AA patient for a PNH clone (flow for CD55/CD59 loss).

BOARD TRAP

A PNH clone in AA does not necessarily mean clinical hemolytic PNH — but it supports the immune pathogenesis (GPI-deficient cells escape T-cell attack) and must be monitored for clonal evolution and thrombosis.

13. PIG-A anchor

WHAT YOU SEE

Anchor labeled PIG at lower right of the hotel

WHAT IT ENCODES

The PNH clone arises from an acquired somatic PIGA mutation (X-linked) in a hematopoietic stem cell, which abolishes the GPI anchor and therefore the complement regulators CD55 and CD59, causing complement-mediated intravascular hemolysis.

BOARD TRAP

PIGA mutation removes GPI-anchored CD55/CD59 → unchecked complement → intravascular hemolysis, thrombosis, and cytopenias. PNH, AA, and MDS form a clonal-marrow-failure spectrum that can evolve into one another.